



Safety Data Sheet

Issue Date: 07-Sep-2022

Revision Date: 18-Feb-2026

Version 3

1. IDENTIFICATION

Product identifier

Product Name Bromoxynil 240 + MCPA 240 EC

Other means of identification

SDS # ADAMA-335

Registration Number(s) EPA Reg. No. 66222-299
UN/ID No NA1993

Recommended use of the chemical and restrictions on use

Recommended use EPA registered pesticide.

Details of the supplier of the safety data sheet

Manufacturer Address

Makhteshim Agan of North America, Inc. (d/b/a ADAMA)
8601 Six Forks Rd., Suite 300
Raleigh, NC 27615
1-919-256-9300

Emergency telephone number

Emergency Telephone For fire, spill and/or leak contact INFOTRAC:
1-800-535-5053 (North America) 1-352-323-3500 (International)
For medical emergencies and health/safety inquiries, contact ProPharma Group:
1-877-250-9291

2. HAZARDS IDENTIFICATION

Appearance Amber liquid

Physical state Liquid

Odor Mild hydrocarbon

Classification

Flammable liquids	Category 4
Acute toxicity - Oral	Category 4
Acute toxicity - Inhalation (Dusts/Mists)	Category 4
Reproductive toxicity	Category 2
Aspiration hazard	Category 1

Label elements



Signal word

Danger

Hazard statements

Combustible liquid.

Harmful if swallowed.

Harmful if inhaled.

Suspected of damaging fertility or the unborn child.

May be fatal if swallowed and enters airways.

Precautionary Statements - Prevention

Obtain special instructions before use.

Do not handle until all safety precautions have been read and understood.

Wear protective gloves, protective clothing, eye protection and face protection.

Wash face, hands and any exposed skin thoroughly after handling.

Avoid breathing dust, fume, gas, mist, vapors and spray.

Do not eat, drink or smoke when using this product.

Use only outdoors or in a well-ventilated area.

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

Precautionary Statements - Response

IF exposed or concerned: Get medical advice/attention.

IF INHALED: Remove person to fresh air and keep comfortable for breathing.

Call a POISON CENTER or doctor/physician if you feel unwell.

IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician.

Do NOT induce vomiting.

Rinse mouth.

Precautionary Statements - Storage

Store in a well-ventilated place. Keep container tightly closed.

Store locked up.

Other Information

Toxic to aquatic life with long lasting effects.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name	CAS No.	Weight-%
MCPA, 2-ethylexyl ester	29450-45-1	35.37
Bromoxynil octanoate (octanoic acid, 2,6-dibromo-4-cyanophenyl ester)	1689-99-2	32-34
Solvent	Proprietary	21-23
Emulsifier 1	Proprietary	5.5-6.5
Emulsifier 2	Proprietary	0.5-1.5
Surfactant	Proprietary	0.5-1.5

If Chemical Name/CAS No is "proprietary" and/or Weight-% is listed as a range, the specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. FIRST AID MEASURES**Description of first aid measures****General Advice**

In case of accident or if you feel unwell, seek medical advice immediately (show the label or SDS where possible).

Eye Contact	Rinse thoroughly with plenty of water, also under the eyelids. If eye irritation persists: Get medical advice/attention.
Skin Contact	Wash with soap and water. If skin irritation persists, call a physician.
Inhalation	IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a physician if you feel unwell.
Ingestion	IF SWALLOWED: call a poison control center or physician immediately. Rinse mouth. Do NOT induce vomiting.

Most important symptoms and effects, both acute and delayed

Symptoms	Harmful if swallowed. Harmful if inhaled. May cause drowsiness or dizziness. May be fatal if swallowed and enters airways.
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Indication of any immediate medical attention and special treatment needed

Notes to Physician	This product contains a petroleum distillate. DO NOT INDUCE VOMITING. Vomiting may cause aspiration pneumonia. Treat symptomatically for ingestion and/or skin and eye contact.
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5. FIRE-FIGHTING MEASURES**Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Carbon dioxide, dry chemical powder, or appropriate foam.

Unsuitable Extinguishing Media Not determined.

Specific Hazards Arising from the Chemical

Combustible liquid and vapor. USE WATER WITH CAUTION. Material will float and may ignite on surface of water.

Hazardous combustion products Smoke, fumes or vapors, and oxides of carbon.

Protective equipment and precautions for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

6. ACCIDENTAL RELEASE MEASURES**Personal precautions, protective equipment and emergency procedures**

Personal Precautions	Use personal protective equipment as required. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area).
For Emergency Responders	Follow all fire fighting procedures in Section 5. Use personal protection recommended in Section 8.

Environmental precautions

Environmental precautions	See Section 12 for additional Ecological Information. See Section 13: DISPOSAL CONSIDERATIONS.
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Methods and material for containment and cleaning up

Methods for Containment	Prevent further leakage or spillage if safe to do so. Prevent run-off to sewers, streams or other bodies of water.
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Methods for Clean-Up

Soak up with inert absorbent material. Use clean non-sparking tools to collect absorbed material. Keep in suitable, closed containers for disposal. For waste disposal, see section 13 of the SDS.

7. HANDLING AND STORAGE

Precautions for safe handling**Advice on Safe Handling**

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Wear appropriate personal protective equipment. Wash face, hands and any exposed skin thoroughly after handling. Avoid breathing dust/fume/gas/mist/vapors/spray. Use only outdoors or in a well-ventilated area. Keep away from heat/sparks/open flames/hot surfaces. — No smoking. Keep container tightly closed.

Conditions for safe storage, including any incompatibilities**Storage Conditions**

Keep containers tightly closed in a cool, well-ventilated place. Store locked up.

Incompatible Materials

Strong oxidizing agents.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure Guidelines

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

Chemical name	ACGIH TLV	OSHA PEL	NIOSH
Surfactant	TWA: 5 ppm	-	-

Appropriate engineering controls**Engineering Controls**

Apply technical measures to comply with the occupational exposure limits.

Individual protection measures, such as personal protective equipment**Eye/Face Protection**

Refer to 29 CFR 1910.133 for eye and face protection regulations.

Skin and Body Protection

Refer to 29 CFR 1910.138 for appropriate skin and body protection.

Respiratory Protection

Refer to 29 CFR 1910.134 for respiratory protection requirements.

General Hygiene Considerations Handle in accordance with good industrial hygiene and safety practice.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical state
Appearance
Color

Liquid
Amber liquid
Amber

Odor
Odor Threshold

Mild hydrocarbon
Not determined

Property**Values****Remarks • Method**

pH

3.95 (1% solution)

Melting point / freezing point

No data available

Initial boiling point and boiling

No data available

range		
Flash point	74 °C / 165 °F	
<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Evaporation rate	Not determined	
Flammability (Solid, Gas)	Not determined	
Flammability Limit in Air		
Upper flammability or explosive limits	No data available	
Lower flammability or explosive limits	No data available	
Vapor Pressure	Not determined	
Relative vapor density	No data available	
Relative Density	1.118	
Water Solubility	Soluble in water	
Solubility in other solvents	Not determined	
Partition Coefficient	Not determined	
Autoignition temperature	No data available	
Decomposition temperature	Not determined	
Kinematic viscosity	16.227 @20C	
Dynamic viscosity	Not determined	
Particle characteristics	No data available	
Explosive Properties	Not determined	
Oxidizing Properties	Not determined	

10. STABILITY AND REACTIVITY

Reactivity

Not reactive under normal conditions.

Chemical stability

Stable under recommended storage conditions.

Possibility of hazardous reactions

None under normal processing.

Hazardous Polymerization

Hazardous polymerization does not occur.

Conditions to Avoid

Excessive heat and fire. Incompatible Materials.

Incompatible materials

Strong oxidizing agents.

Hazardous decomposition products

Smoke, fumes or vapors, and oxides of carbon.

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure

Product Information

Eye Contact Avoid contact with eyes.

Skin Contact Avoid contact with skin.

Inhalation Harmful if inhaled.

Ingestion

Harmful if swallowed.

Component Information

Chemical name	Oral LD50	Dermal LD50	Inhalation LC50
MCPA, 2-ethylexyl ester 29450-45-1	= 1300 mg/kg (Rat)	> 2000 mg/kg (Rat)	> 5.05 mg/L (Rat) 4 h
Bromoxynil octanoate (octanoic acid, 2,6-dibromo-4-cyanophenyl ester) 1689-99-2	= 238 mg/kg (Rat)	> 2000 mg/kg (Rat)	= 0.721 mg/L (Rat) 4 h = 0.809 mg/L (Rat) 4 h
Solvent	> 5000 mg/kg (Rat)	> 2000 mg/kg (Rabbit)	> 590 mg/m ³ (Rat) 4 h
Emulsifier 1	-	> 2000 mg/kg (Rat)	-
Emulsifier 2	1086 - 1980 mg/kg (Rat)	-	-
Surfactant	= 3730 mg/kg (Rat)	= 1980 mg/kg (Rabbit)	> 227 ppm (Rat) 6 h

Symptoms related to the physical, chemical and toxicological characteristics**Symptoms**

Please see section 4 of this SDS for symptoms.

Delayed and immediate effects as well as chronic effects from short and long-term exposure**Carcinogenicity**

Based on the information provided, this product does not contain any carcinogens or potential carcinogens as listed by OSHA, IARC or NTP.

Reproductive toxicity

Suspected of damaging fertility or the unborn child.

Aspiration hazard

May be fatal if swallowed and enters airways.

Numerical measures of toxicity

ATEmix (oral) <2,000 mg/kg

ATEmix (dermal) >2,000 mg/kg

ATEmix (inhalation-dust/mist) 4.82 mg/kg

12. ECOLOGICAL INFORMATION**Ecotoxicity**

Toxic to aquatic life with long lasting effects.

Component Information

Chemical name	Algae/aquatic plants	Fish	Crustacea
MCPA, 2-ethylexyl ester 29450-45-1	EC50: =0.46mg/L (72h, Pseudokirchneriella subcapitata) EC50: =0.43mg/L (96h, Pseudokirchneriella subcapitata)	LC50: 3.2 - 4.6mg/L (96h, Lepomis macrochirus) LC50: =3.2mg/L (96h, Oncorhynchus mykiss) LC50: >0.55mg/L (96h, Lepomis macrochirus)	EC50: =0.29mg/L (48h, Daphnia magna)
Solvent		LC50: =19mg/L (96h, Pimephales promelas) LC50: =2.34mg/L (96h, Oncorhynchus mykiss) LC50: =1740mg/L (96h, Lepomis macrochirus) LC50: =45mg/L (96h, Pimephales promelas)	EC50: =0.95mg/L (48h, Daphnia magna)

		LC50: =41mg/L (96h, Pimephales promelas)	
Chemical name	Algae/aquatic plants	Fish	Crustacea
Emulsifier 1		LC50: >45mg/L (96h, Danio rerio)	
Emulsifier 2		LC50: =10.8mg/L (96h, Oncorhynchus mykiss)	
Surfactant	EC50: =11.5mg/L (72h, Desmodium subspicatus)	LC50: 32 - 37mg/L (96h, Oncorhynchus mykiss) LC50: >7.5mg/L (96h, Oncorhynchus mykiss) LC50: 27 - 29.5mg/L (96h, Pimephales promelas) LC50: =29.7mg/L (96h, Pimephales promelas) LC50: 10.0 - 33.0mg/L (96h, Lepomis macrochirus)	EC50: =39mg/L (48h, Daphnia magna)

Persistence/Degradability

Not determined.

Bioaccumulation

There is no data for this product.

Mobility

Chemical name	Partition coefficient
MCPA, 2-ethylexyl ester 29450-45-1	4.49
Solvent	6.5
Emulsifier 1	4.297
Surfactant	2.9

Other adverse effects

No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected.

13. DISPOSAL CONSIDERATIONS

Disposal methods**Disposal of Wastes**

Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging

Disposal should be in accordance with applicable regional, national and local laws and regulations.

14. TRANSPORT INFORMATION

Note

Please see current shipping paper for most up to date shipping information, including exemptions and special circumstances.

DOT

NOT REGULATED if shipped in NON BULK packaging (single containers less than 119 gal/ 882 lbs) by ground transport
 BULK Packaging (single containers larger than 119 gal/ 882 lbs):
 UN/ID No NA1993
 Proper Shipping Name Combustible liquid, n.o.s. (Petroleum distillates, Bromoxynil)
 Transport hazard class(es) Comb liq
 Packing Group III
 Marine Pollutant This material ships as a marine pollutant when single container exceeds 119 gal/ 882 lbs.

IATA

UN number or ID number UN3082
 Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s. (Bromoxynil)
 Transport hazard class(es) 9
 Packing group III
 Description This material ships as a marine pollutant when inner packagings exceed 5L/5KG

IMDG

UN number or ID number UN3082
 Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s. (Bromoxynil)
 Transport hazard class(es) 9
 Packing Group III
 Marine Pollutant This material ships as a marine pollutant when inner packagings exceed 5L/5KG

15. REGULATORY INFORMATION

International Inventories

Chemical name	TSCA	TSCA Inventory Status	DSL/NDSL	EINECS/ELINCS	ENCS	IECSC	KECL	PICCS	AICS
MCPA, 2-ethylexyl ester				X		X			
Bromoxynil octanoate (octanoic acid, 2,6-dibromo-4-cyanophenyl ester)	X	ACTIVE	X	X		X			
Solvent	X	ACTIVE	X	X		X	X	X	X
Emulsifier 1	X	ACTIVE	X	X	X	X	X	X	X
Emulsifier 2	X	ACTIVE	X	X	X	X	X	X	X
Surfactant	X	ACTIVE	X	X	X	X	X	X	X

Legend:

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
EINECS/ELINCS - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances
ENCS - Japan Existing and New Chemical Substances
IECSC - China Inventory of Existing Chemical Substances
KECL - Korean Existing Chemicals Inventory
PICCS - Philippines Inventory of Chemicals and Chemical Substances
AICS - Australian Inventory of Chemical Substances

US Federal Regulations**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

Chemical name	Hazardous Substances RQs	Extremely Hazardous Substances RQs	Reportable Quantity (RQ)
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Emulsifier 2	1000 lb / 54 kg (final RQ)		RQ 1000 lb final RQ RQ 454 kg final RQ
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SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

Chemical name	CAS No.	Weight-%	SARA 313 - Threshold Values %
Bromoxynil octanoate (octanoic acid,2,6-dibromo-4-cyanophenyl ester) - 1689-99-2	1689-99-2	32-34	1.0

CWA (Clean Water Act)

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42)

Chemical name	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants	CWA - Hazardous Substances
Emulsifier 2	1000 lb			X

US State Regulations**California Proposition 65**

This product contains the following Proposition 65 chemicals:

Chemical name	California Proposition 65
Bromoxynil octanoate (octanoic acid,2,6-dibromo-4-cyanophenyl ester) - 1689-99-2	Developmental

U.S. State Right-to-Know Regulations

This product contains the following State Right-to-Know chemicals:

Chemical name	New Jersey	Massachusetts	Pennsylvania
Bromoxynil octanoate (octanoic acid,2,6-dibromo-4-cyanophenyl ester) 1689-99-2	X		
Emulsifier 2	X	X	X
Surfactant		X	X

EPA Pesticide Registration Number EPA Reg. No. 66222-299**EPA Statement**

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. Following is the hazard information as required on the pesticide label:

EPA Pesticide Label

Please refer to EPA label for additional information

Difference between SDS and EPA pesticide label

Please refer to EPA label for additional information

16. OTHER INFORMATION**NFPA**
HMISHealth hazards -
Health hazards -Flammability -
Flammability -Instability -
Physical hazards -Special hazards -
Personal protection -

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Revision Note: SDS sections updated

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet